

Accident Hotspot Identification Dashboard/Smart AccidentPredict

Team No. 15

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DISCOVER YOUR WORLD

Introduction (Iteration 1)

Project Overview

- **Purpose:** Identify and visualize driving incident hotspots in Breda (speeding, abrupt braking, sharp turns, rapid acceleration) using clustering algorithms and machine learning.
- **Method:** Analyze historical incident data, label clusters by predominant incident types, and predict labels on unseen data.
- **Objective:** Develop a dashboard for proactive resource allocation and safety measures implementation...

Relevance and Interest

- Improved road safety and reduction of accidents.
- Saved lives and improved quality of life in Breda.
- Proactively allocated resources by city planners and implemented safety measures by the police and authorities.

Significance of Machine Learning

- Extract meaningful patterns from complex datasets.
- Clustering identifies high incidence areas.
- Predictive models forecast future hotspots.
- Enables accurate interventions for a safer driving environment.

Introduction (Iteration 2)

Project Overview

- **Purpose:** Develop an advanced dashboard integrating a machine learning model to predict the severity of traffic accidents based on specific accident details. Identify dangerous road intersections using unsupervised clustering.
- **Method:** Analyze real-time data from cameras and sensors, use clustering techniques to identify and analyze the most dangerous intersections in Breda, and predict accident severity using machine learning and deep learning models.
- **Objective:** Enhance road safety in Breda by providing actionable insights, enabling efficient mobilization of law enforcement and emergency services, and informing city planners and traffic management authorities.

Relevance and Interest

- **Improved Road Safety:** Address the critical need to improve road safety and reduce accident-related fatalities and injuries.
- **Quality of Life:** Potentially save lives and enhance the overall well-being and quality of life in Breda.
- **Efficient Resource Allocation:** Enable city planners to proactively allocate resources and authorities to implement timely safety measures.

Significance of Machine Learning

- **Data Analysis:** Analyze large and complex datasets to uncover patterns not immediately apparent through traditional methods.
- **Real-time Monitoring:** Process real-time data from cameras and sensors for continuous monitoring and predictive capabilities.
- **Predictive Modelling:** Create models that forecast the severity of accidents and identify high-risk intersections.
- **Proactive Safety Measures:** Ensure safety measures are both responsive and preventative, leading to a safer driving environment in Breda.

Problem Statement (Iteration 1)

Problem Definition

- Objective: Identify and predict hotspots of unsafe driving behaviors in Breda.
- Behaviors: Speeding, abrupt braking, sharp turns, rapid acceleration.
- Approach: Utilize machine learning to analyze historical incident data, cluster high-incidence areas, and predict future hotspots.

Context and Background

- Road Safety Concerns: Unsafe driving behaviours contribute to accidents, injuries, and property damage.
- Breda's Scenario: Urban area facing various driving incidents compromising road safety.
- Data Source: ANWB Safe Driving App provides incident data, but lacks predictive analysis.

Importance and Benefits

- Road Safety Enhancement: Crucial for reducing accident rates and improving safety.
- Strategic Resource Allocation: Enables police and city planners to deploy resources effectively.
- Targeted Interventions: Implementation of measures like speed cameras for proactive safety measures.
- Public Safety Optimization: Enhances safety for drivers and pedestrians while optimizing resource utilization.

Problem Statement (Iteration 2)

Problem Definition

- **Objective:** Develop a predictive model to assess and mitigate the severity of traffic accidents in Breda using machine learning techniques.
- **Challenges:** Law enforcement and emergency services often arrive too late at accident scenes, affecting service quality and community safety. Delayed responses can be life-threatening in severe incidents.
- **Approach:** Utilize real-time data from cameras and sensors deployed across the city to classify accidents into categories of severity (Fatal, Injured, Material damage only) based on factors like road details, date of occurrence, weather conditions, and vehicle characteristics.

Context and Background

- **Road Safety Concerns:** Despite efforts by organizations like ANWB, traffic accidents remain a significant risk to public health and safety in Breda. The city's investments in road infrastructure and safety campaigns have not fully mitigated these issues.
- **Breda's Scenario:** A bustling urban area facing ongoing challenges related to traffic management and accident prevention.
- **Current Methods:** Existing methods of accident response and prevention can be enhanced through predictive analytics to enable more effective resource deployment and targeted safety measures.

Importance and Benefits

- **Road Safety Enhancement:** Machine learning-driven predictions can significantly reduce accident rates and improve overall road safety.
- **Strategic Resource Allocation:** Enables efficient and proactive deployment of law enforcement and emergency services.
- **Targeted Interventions:** Provides actionable insights for implementing measures like speed cameras and improving high-risk intersections.
- **Public Safety Optimization:** Enhances safety for drivers and pedestrians, leading to cost savings in healthcare, vehicle repairs, and insurance claims, while improving the quality of life in Breda.

AI Project Canvas

Title: **Accident Hotspot Identification Dashboard**

Data  <i>Which data do you need?</i> • ANWB (Safe Driving) Data - driving incidents, including speeding, braking, turns, acceleration • KNMI Weather Data - precipitation, temperature, and wind conditions • BRON (Accidents) Data - traffic accidents, including the date, time, location, severity, and conditions • Argaleo Greenery and Road Data	Skills  <i>Which skills do you need for development?</i> • Development Skills • Communication Skills • Project Management • Infrastructure Knowledge • Machine Learning • Data Science and Analytics • Visualization and Programming Output  <i>Which key metric are you optimizing for?</i> • Accurate data • Fast delivery • Transparent output • Clustering Accuracy • Prediction Accuracy • User Satisfaction	Value Proposition  <i>What is the value added by your project?</i> • Actionable Road Insights: Providing actionable insights into accident hotspots to help law enforcement and city planners proactively prevent incidents. • Increased Efficiency: Giving our end users the best insight for efficient road navigation. • Improved safety: By providing road safety and risk statistics we increase the likeliness off a save trip for our end users.	Integration  <i>How will the project be integrated?</i> • Police/Authorities Systems • Interactive map/dashboard with Folium or Plotly • Database Connectivity Stakeholders  <i>Who are the key stakeholders?</i> • ANWB • Police Department • Breda Municipality • Traffic Management Authorities • Data Providers	Customers  <i>Who are the end customers?</i> • Police Officers and Traffic Enforcement Units: Utilize the dashboard for strategic resource allocation. • Breda Municipality Officials: Use insights for policy decisions and infrastructure improvements. • Citizens of Breda: Indirect beneficiaries who experience increased road safety.
Cost  <i>What costs will the project incur?</i> • Cloud Services: Expenses for cloud storage, server uptime, computing resources, and database management. • Software and Tools: Licenses for development tools, libraries, and software. • Maintenance: Ongoing costs for system maintenance, updates, and technical support. • Training and Workshops: Expenses for training police officers and stakeholders to use the system effectively.		Revenue  <i>How will the project generate revenue?</i> • Subscription Model • Consulting Services • Licensing Fees • Service Contracts • Data Monetization • Advertisement and Sponsorship • Grants and Government Funding		

Data Description (Iteration 1)

Datasets

- ANWB (Safe Driving) Data – records of driving incidents (speeding, harsh cornering, hard acceleration, and abrupt braking)
- KNMI Weather Data – historical weather data (precipitation, temperature, and wind conditions)
- BRON (Accidents) Data - details of traffic accidents (date, time, location, severity, and conditions at the time of the accident)
- Argaleo Greenery and Road Data - information on the greenery and road infrastructure in Breda

Attributes/features

- Utilize all available variables in the dataset for initial analysis.
- Employ unsupervised learning methods like clustering.
- Uncover inherent structures and patterns in the data without bias towards specific attributes.
- Determine the significance of each feature based on their contribution to forming distinct clusters.

Data Description (Iteration 2)

Datasets

- ANWB (Safe Driving) Data – records of driving incidents (speeding, harsh cornering, hard acceleration, and abrupt braking)
- KNMI Weather Data – historical weather data (precipitation, temperature, and wind conditions)
- BRON (Accidents) Data - details of traffic accidents (date, time, location, severity, and conditions at the time of the accident)
- "Accidents_17_23" dataset with comprehensive information on accidents and their severity.
- Argaleo Greenery and Road Data - information on the greenery and road infrastructure in Breda

Attributes/features

- Utilize all available variables in the datasets for initial analysis.
- Employ unsupervised learning methods like clustering to uncover inherent structures and patterns in the data.
- Determine the significance of each feature based on their contribution to forming distinct clusters.
- Focus on the Accidents_17_23 dataset for accident severity prediction due to its comprehensive information.

Data Description (Iteration 1)

Data Cleaning and Pre-processing

- Handling missing values, removing duplicates and invalid data, standardizing units of measurement, and ensuring consistency in data formats (date, time, other measurements).
- Merging datasets based on common keys, such as location coordinates and timestamps.
- Feature engineering may be required to create new variables that capture complex interactions between existing features, such as combining weather conditions with driving behaviors to predict accident severity.

Data Description (Iteration 2)

Data Cleaning and Pre-processing

- Handling missing values
- Address significant missing values, particularly in the original BRON dataset, by focusing on the more comprehensive Accidents_17_23 dataset
- Standardizing units of measurement
- Ensure consistency in data formats, such as date, time, and other measurements across all datasets
- Ensuring Consistency in Data Formats:
- Standardize formats for seamless integration and analysis.

Methodology (Iteration 1)

Approach

Comprehensive machine learning workflow integrating clustering and predictive modeling to analyze and forecast accident-prone areas based on various data attributes.

Combining historical accident data, provided weather information, and road characteristics.

Methodology (Iteration 2)

Approach

Comprehensive Workflow:

Integrating clustering and predictive modelling to analyze and forecast accident-prone areas and accident severity.

Data Collection:

Primary datasets from ANWB, focusing on traffic incidents and accidents, driver behaviour, and environmental conditions.

Additional data from KNMI, BRON, and Argaleo for weather and road characteristics.

Data Pre-processing:

Handle missing values, remove duplicates, standardize measurements, and ensure consistency.

Split data into training, validation, and test sets.

Feature Engineering:

Extract relevant features from datasets.

Develop new variables to capture complex interactions.

Model Selection:

Supervised learning for accident severity prediction (Logistic Regression, Random Forest, Gradient Boosting, CNN).

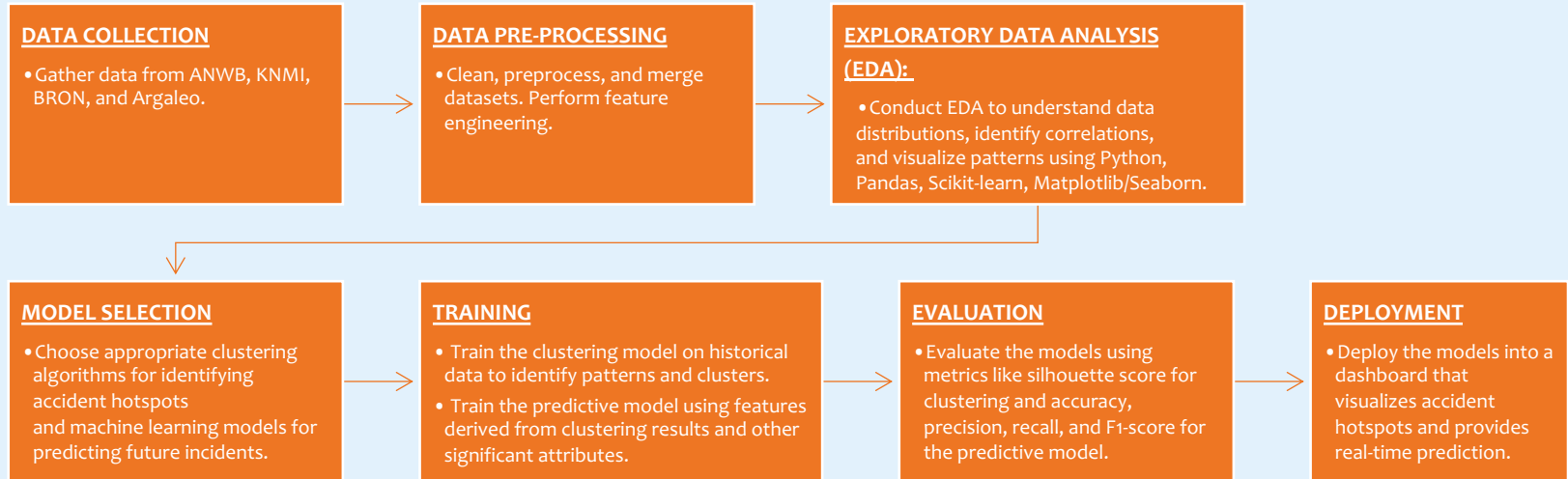
Unsupervised learning for intersection clustering (K-means, DBSCAN).

Model Deployment:

Integrate the best model into a Streamlit dashboard for real-time predictions and insights.

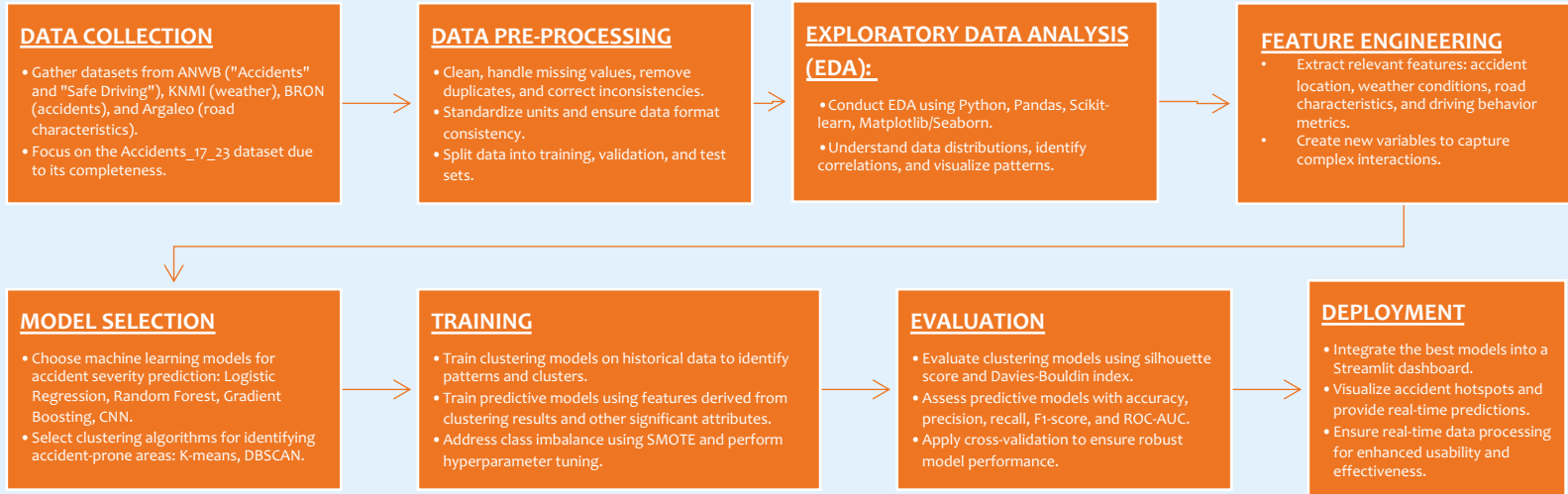
Methodology (Iteration 1)

Steps



Methodology (Iteration 2)

Steps



Methodology (Iteration 1)

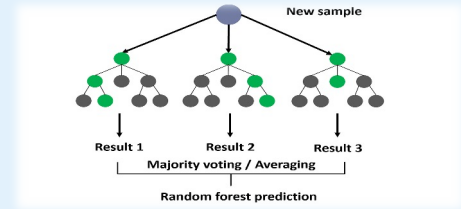
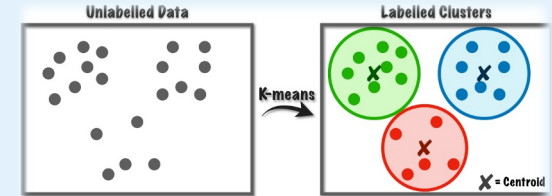
Models and Their Architecture

Clustering Model:

- **Algorithm:** K-means clustering
- **Reason:** K-means is effective for identifying distinct groups in large datasets based on similarity. It is suitable for spatial clustering to locate accident hotspots.
- **Architecture:** The model uses features like latitude, longitude, time of day, and weather conditions as input to cluster incident locations. The output is a set of clusters representing accident hotspots.

Predictive Model:

- **Algorithm:** Random Forest or Gradient Boosting Machine (GBM)
- **Reason:** Both algorithms are robust for handling complex, non-linear relationships and provide high accuracy and interpretability for classification tasks.
- **Architecture:** The model uses input features such as historical accident data, weather data, and road characteristics to predict the likelihood of accidents in specific areas. The output is the probability of an accident occurring in each identified hotspot.



Methodology (Iteration 2)

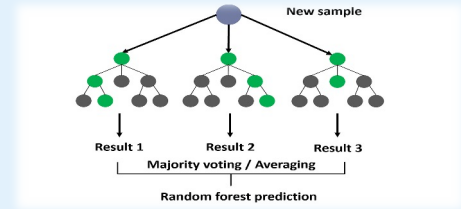
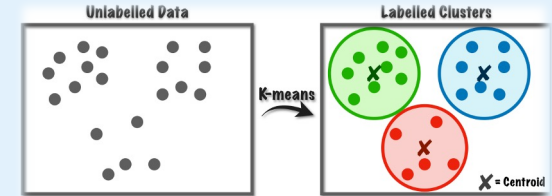
Models and Their Architecture

Clustering Model:

- **Algorithm:** K-means Clustering and DBSCAN
- **Reason:** K-means is effective for identifying distinct groups in large datasets based on similarity. DBSCAN is useful for detecting clusters of varying shapes and densities. Both are suitable for spatial clustering to locate accident hotspots.
- **Architecture:**
 - **Input Features:** Latitude, longitude, time of day, and weather conditions.
 - **Output:** Clusters representing accident hotspots.

Predictive Model:

- **Algorithm:** Logistic Regression, Random Forest, Gradient Boosting, and Convolutional Neural Network (CNN)
- **Reason:**
 - **Random Forest and Gradient Boosting:** Robust for handling complex, non-linear relationships, providing high accuracy and interpretability for classification tasks.
 - **CNN:** Effective for capturing spatial patterns and geographical distribution of accidents.
- **Architecture:**
 - **Input Features:** Historical accident data, weather data, road characteristics, and driving behaviour metrics.
 - **Output:** Probability of an accident occurring in each identified hotspot.



Risks and Legal Obligations (Iteration 1)

Risk Associated with the AI System

↓

Data Quality: Inaccurate or incomplete data could lead to poor model performance.

↓

Model Accuracy: Incorrect predictions could result in inefficient resource allocation and potential safety risks.

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Bias: The models might inadvertently learn and propagate biases present in the historical data.

Legal Obligations and Approach

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Data Privacy: Ensure compliance with GDPR by anonymizing personal data and obtaining necessary consents for data usage.

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Transparency: Maintain transparency in how data is collected, processed, and used. Provide clear documentation of model workings and decision-making processes.

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Ethical Use: Ensure the system is used ethically, particularly in avoiding discriminatory practices and ensuring public safety

Risks and Legal Obligations (Iteration 2)

Risk Associated with the AI System



Data Quality: Inaccurate or incomplete data could lead to poor model performance.



Model Accuracy: Incorrect predictions could result in inefficient resource allocation and potential safety risks.



Bias: The models might inadvertently learn and propagate biases present in the historical data.

Legal Obligations and Approach:

High-Risk Classification (EU AI Act Articles 6-15):

Reason: Predicts incidents influencing safety measures and law enforcement actions.

Transparency (Article 52):

Inform users that predictions are AI-generated to foster trust and understanding.

Post-Market Monitoring (Articles 61-62):

Continuous monitoring to ensure ongoing compliance, safety, and reliability after deployment. Regular updates to maintain accuracy and prevent bias.

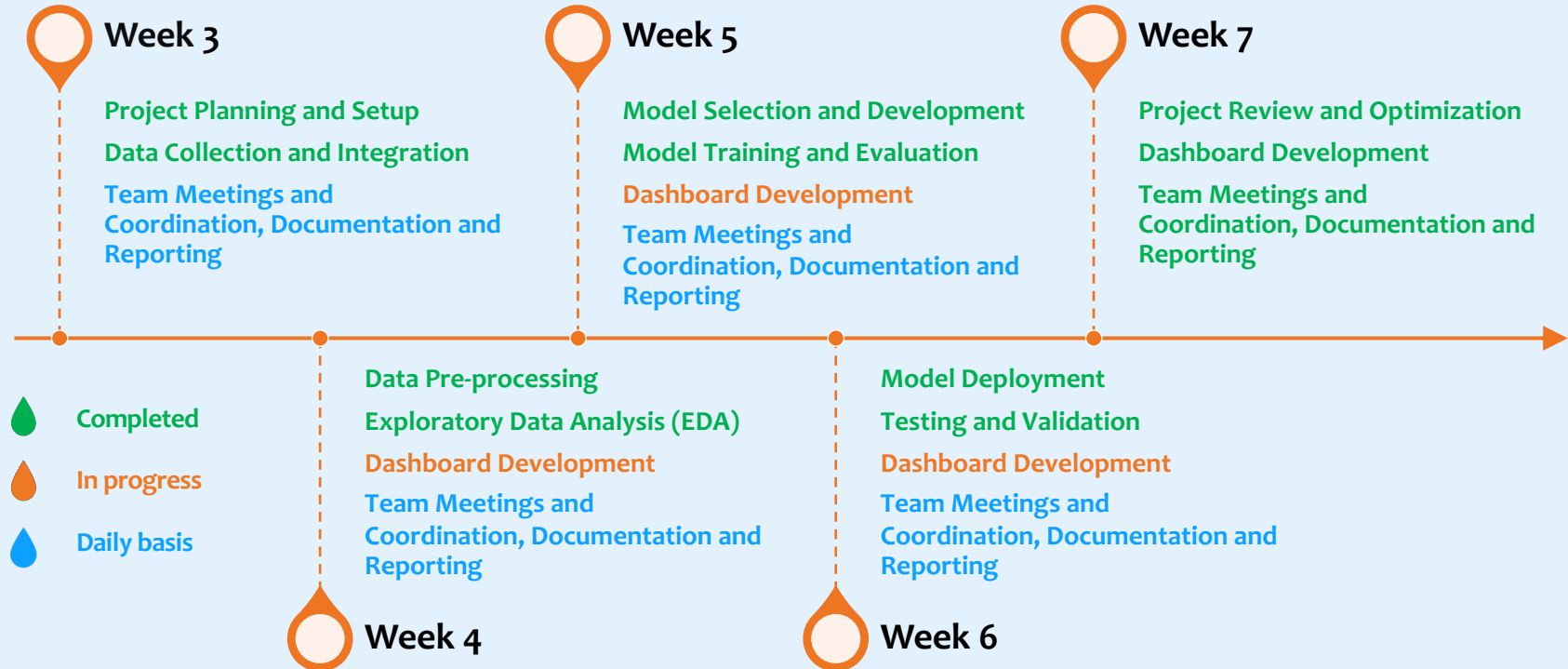
Prohibited Practices (Article 5):

Avoid exploiting vulnerabilities, using subliminal techniques, or engaging in social scoring.

Compliance Measures:

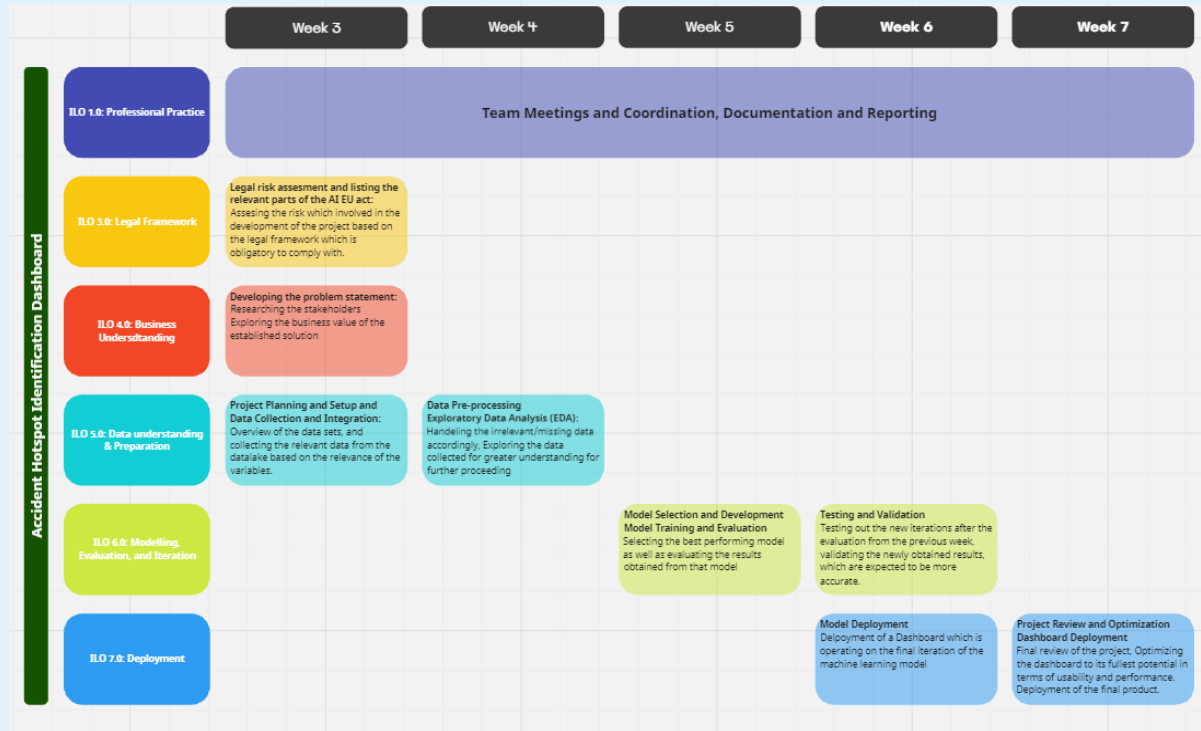
- Maintain detailed documentation of AI system development, data sources, and decision-making processes.
 - Provide training to users on AI system capabilities and limitations.
- Implement continuous monitoring to regularly update and assess the system for accuracy, bias, and potential harm.

Project Timeline



Project Roadmap

[Link to the roadmap](#)



References

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Thank you for your attention!

